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Part 3 – Test Plan for Weather station focusing Station status monitoring unit(W5)

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Weather station – Station status monitoring unit(W5)

3 Test Plan and Rework

3.1 Rework on former submissions

After reviewing Part 1 and Part 2, no rework was necessary. The requirements, status evaluation logic (GREEN/YELLOW/RED), and priority rules remain consistent. Interfaces between status calculation, maintenance, and user interface components are unchanged.

3.2 Test Plan

3.2.1 Component Test

Component testing ends when full branch coverage is achieved and all Green/Yellow/Red paths are validated without open critical defects.

Why useful?

Because W5 contains decision logic that directly influences maintenance scheduling and dashboard alerts. Any misclassification could lead to wrong operational decisions.

- Explain how the code coverage can be determined.

  Code coverage can be determined using a tool such as JaCoCo (Java Code Coverage Tool). It measures statement coverage, branch coverage, and condition coverage to ensure that all relevant execution paths and decision logic are tested adequately.

- Which static test methods might be useful for your unit?

  Code Review -To verify decision logic correctness,Check threshold values
  Static Code Analysis - Detect unused variables, Detect possible null pointer risks
  Walkthrough - Step-by-step review of algorithm logic

3.2.2 Integration Test

Strategy for Integration Test: A Top-Down Integration Testing strategy is used for the Weather Station system. This strategy is appropriate because the system follows a central control flow starting from the main controller, with higher-level modules coordinating lower level ones. Using Top-Down integration avoids the complexity and fault-isolation issues of a Big Bang approach.

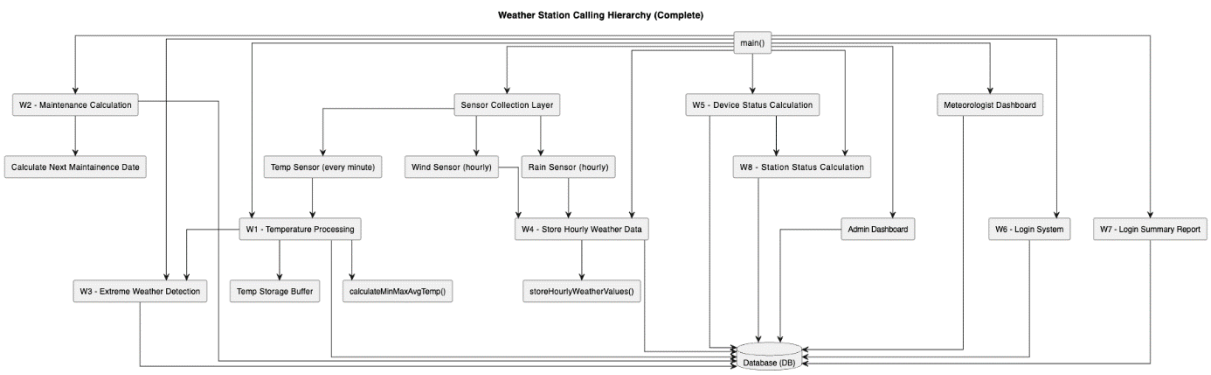

Figure 1: Calling Hierarchy of the Weather Station

The system consists of Sensor, Processing, Maintenance, Storage and User Interface layers. W5 evaluates device status (Green/Yellow/Red), which influences dashboards and maintenance logic.

Describe for your unit, how it is included in the integration testing and which aspects are important for your unit.

In integration testing, W5 is integrated after device health data is available and before station status aggregation (W8) and maintenance handling (W2). Its calculated device states are forwarded to higher-level modules and user-facing components, ensuring that W5 operates correctly within the overall system flow.

Key aspects to focus on for W5 in integration testing

For W5, correct transfer of device data (error rate and age) and proper handling of boundary conditions are essential. Additionally, the accurate propagation of calculated device states to dashboards and database storage is critical to avoid incorrect maintenance actions.

Would it be a good idea to conduct reviews? Which ones? Consider the review methods defined by IEEE1028.

Yes. According to IEEE 1028, a Technical Review and an Inspection are appropriate to verify interface correctness and detect integration defects.

According to IEEE 1028:

1. **Technical Review** - Check interaction logic ,Verify interface contracts
2. **Inspection** - Formal defect detection in integration points

3.2.3 System Test

UML Use Case Model The figure 2 below shows the actors and interactions across the Sensor Layer, Processing Layer, Maintenance Layer, Storage Layer, and User Interface Layer:

The system consists of multiple layers:

- Sensor Layer collects temperature, rainfall, and wind speed values.
- Processing Layer processes weather data (min/max/avg), detects extreme weather days, and evaluates device status (Green/Yellow/Red) and station status.
- Maintenance Layer calculates the next maintenance date, detects overdue/faulty devices, and generates maintenance alerts.
- Storage Layer stores sensor data, error logs/status, and maintenance schedules.
- User Interface Layer allows meteorologists and admin/maintenance staff to log in, view weather reports/statistics, view device status dashboards, and view maintenance notifications.

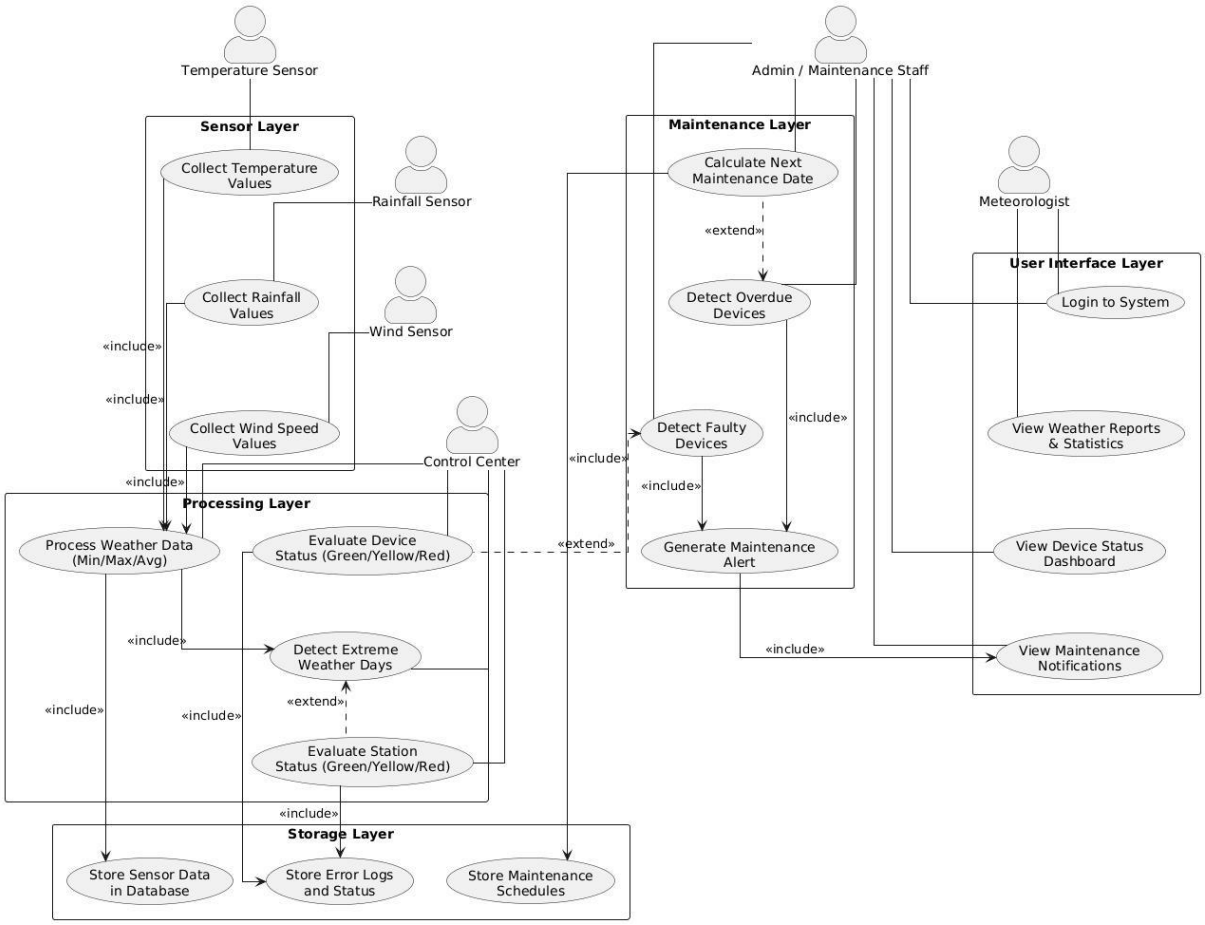

Figure 2: Use-case based UML Diagram of Weather Station

Within the Processing Layer, device-level evaluation is performed by the use case "Evaluate Device Status (Green/Yellow/Red)", which is implemented by Unit W5. The calculated per-device statuses are then stored and used by downstream processes such as maintenance actions and user dashboards, making W5 central to system behavior.

Develop up to 5 test cases (high level) for the system test. One test case should focus your unit. Explain why these test cases are useful.

ID	Scenario	Description	Expected Result	Why Useful
ST-01	Login and View Weather Reports	A meteorologist logs in and opens "View Weather Reports & Statistics".	Login succeeds; correct reports/statistics are displayed.	Verifies end-to-end flow across UI + login + stored weather data (basic system usability).
ST-02	Extreme Weather Detection	Sensor values exceed extreme thresholds and are processed.	The day is flagged as extreme and appears in reports/alerts.	Validates correct processing + interaction with extreme weather logic and reporting.
ST-03 (Focus: W5)	Evaluate Device Status (W5)	The system receives device data where one device crosses a threshold (e.g.,	The affected device status becomes **Yellow/Red** correctly,	Directly validates W5's core responsibility **inside the full system**,

ID	Scenario	Description	Expected Result	Why Useful
		error rate or age becomes critical) while others remain normal.	appears in **Device Status Dashboard**, and is stored in **Error Logs and Status**.	including UI display + DB logging.
ST-04	Maintenance Alert Generation	A device is classified as faulty/critical and maintenance workflow is triggered.	A maintenance alert is generated and visible to admin/maintenance staff; maintenance schedule updated/stored.	Ensures integration between device evaluation and maintenance handling + notifications.
ST-05	Storage Consistency Across Days	Over multiple days, the system collects sensor/device data and users access dashboards/notifications.	Sensor data, device statuses, and maintenance schedules are consistently stored and retrievable; no missing/duplicate records.	Verifies long-running system behavior and traceability across storage + UI + processing.

These high-level test cases validate the most important **end-to-end behaviors** from the UML use-case model: login, reporting, extreme event detection, device status evaluation, maintenance response, and persistence. A dedicated test for **W5** confirms that device classification works correctly not only in isolation, but also when connected to dashboards, storage, and maintenance processes.

Would it be a good idea to conduct reviews? Which ones? Consider the review methods de fined by IEEE1028.

Yes, reviews are recommended at system level.

- **Walkthrough**:
  Test cases are presented to the team to identify missing scenarios or incorrect assumptions about unit interactions before execution.
- **Technical Review**:
  Structured review to ensure coverage of critical system flows, especially interactions between W5, maintenance logic, dashboards, and database storage.

3.2.4 Acceptance Test

Acceptance testing focuses on the customer perspective, especially the accuracy of device classification, real-time updates, reliable maintenance alerts, and correct dashboard display. Particular attention is given to avoiding false critical alerts and ensuring that all status changes are correctly logged. In addition, usability and clarity of status visualization for meteorologists and maintenance staff are important to support operational decision-making.